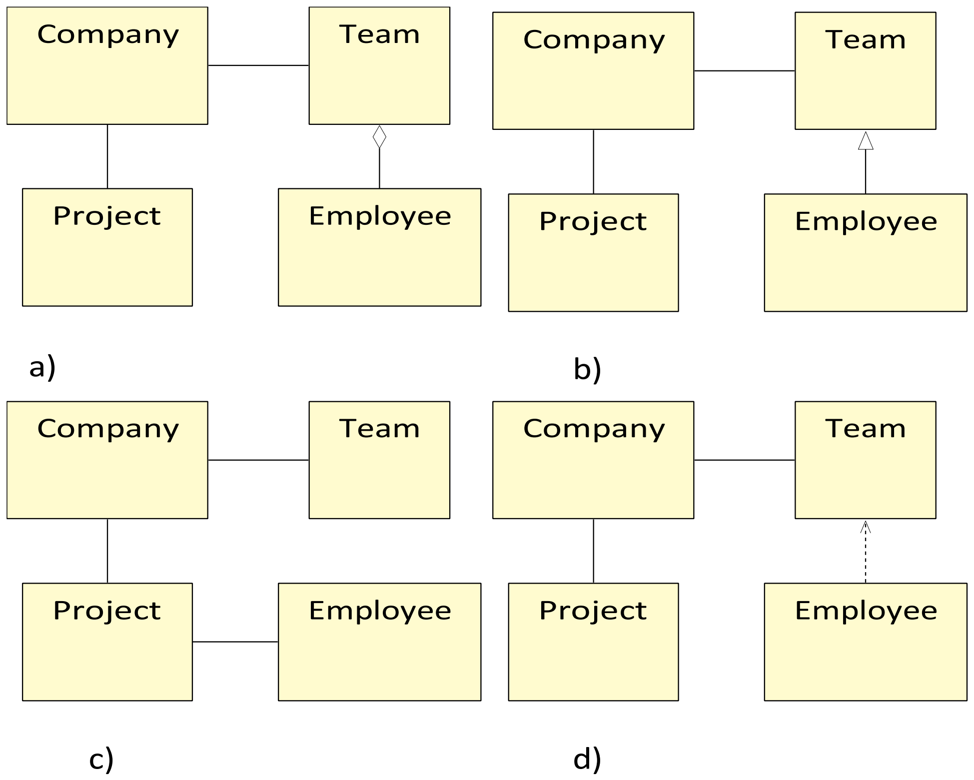
Exercise 1

- Read the following description and select the most appropriate class diagram from adjoining Figure :

"A company realizes projects; each project is executed by a team of employees".

For the sake of simplicity, multiplicities and attributes are not indicated.



Exercise 2

- Consider the following simplified description of a university where professors teach seminars in which students can enrol.

A professors has a name, address, phone number, email address, and salary. A student has also a name, etc., but no salary. A student, however, has an average mark (of the final marks of his or her seminars). A seminar has a name and a number. When a student is enrolled in a seminar, the marks for this enrolment are recorded and the current average as well as the final mark (if there is one) can be obtained from the enrolment. From a student one can obtain a list of seminars he or she is enrolled in. Professors teach seminars. Each seminar has at least one and at most three teachers. There are two types of seminar: bachelor and master. From a bachelor seminar, students can not withdraw. From a master seminar they can.

- Draw a class diagram for this university. Add attributes and methods when necessary. You do not have to include getters and setters for attributes. Visibility modifiers (public, private, etc.) are not mandatory.

- Motivate your decisions and reason them appropriately.
- *Grading policy for this question*
  - Missing motivation) -5 points per important issue.
  - Missing features (methods, attributes)) -3 points per issue.
  - Small syntactic error) -2 points per error.
  - Semantic error (inconsistency, missed feature, etc.)) -5 points per issue.

Exercise 3

- Consider the following required functionality of a following printer service:

The printer service provides customers the possibility to print posters, flyers, or books on demand. The customer should be able to select a type of product (poster, flyer, or book), a desired quantity, and a paper type. In case a book has to be printed, additionally the customer can choose between hard cover and soft cover. Finally, the customer needs to provide a PDF file containing the desired content.

In order for the customer to be able to place an order, he or she must have an account. The customer can create an account by choosing a username/password combination. Furthermore, his or her address and credit card number can be linked to the account, which is required information when placing an order.

Once a customer has provided the information for an order, the system checks if all required information is there, either given in the order (type of product, quantity, etc.), or in the account (address and payment information). If any information is lacking, the system will inform the customer that it needs to be added before the order can be placed. Once all information is in place, the order is placed, and the credit card information is sent to the bank for approval. If the bank approves the card, the order is finalized.

A printing agent is in charge of actually performing the printing. He or she inspects the provided PDF files of finalized orders. If a file does not meet the quality requirements, the customer will be informed about this, and the order is temporarily put on hold until the customer has provided a new PDF file.

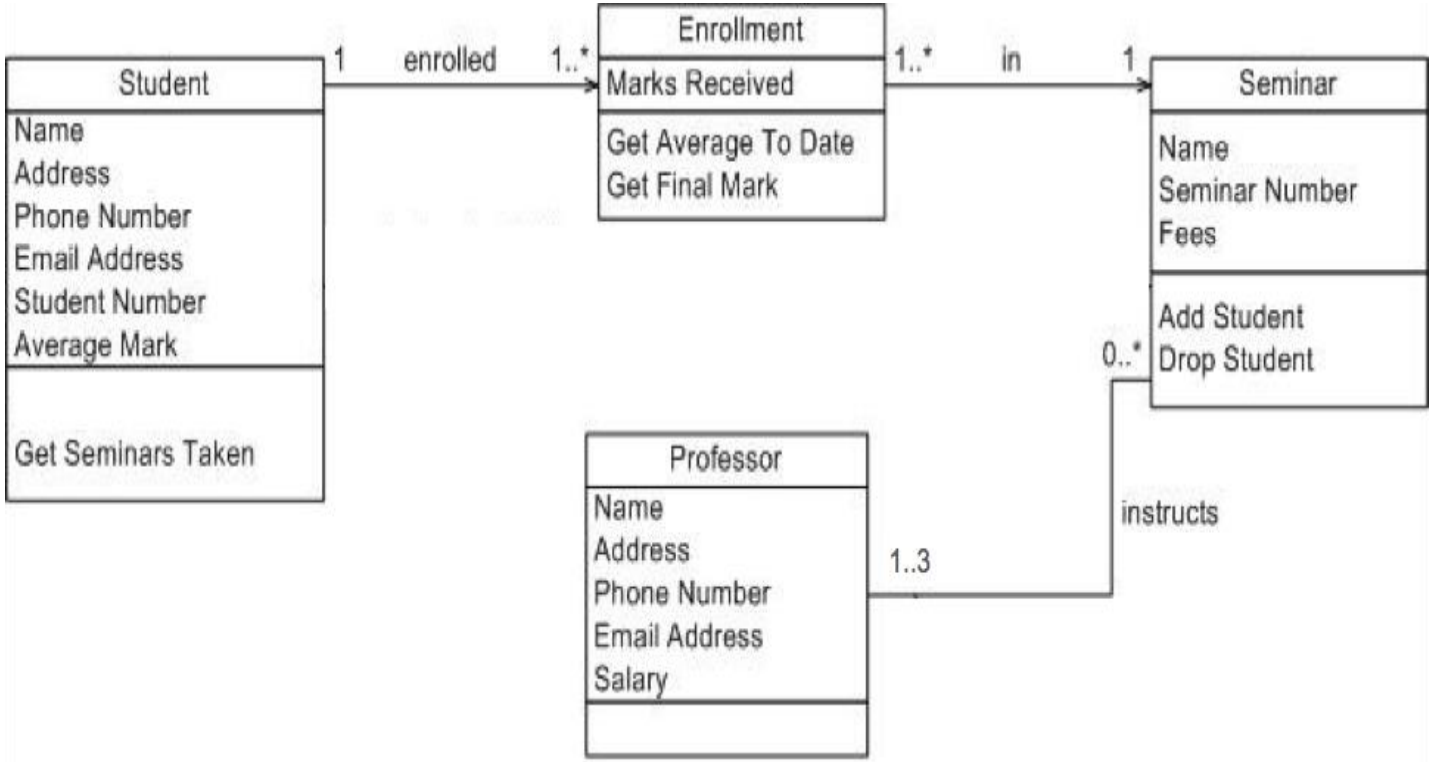
Finally, the administrator monitors if at all times, sufficient paper and ink stock is present. Whenever the amount of paper or ink is running low, an order must be placed at the appropriate supplier (either the paper or ink supplier).

Exercise 3 … cont’d

- Create a UML Use Case Diagram for the printer service. In addition, give a detailed scenario (pre-condition, trigger, guarantee, main scenario, ...) of the use case for “place an order”.
- Based on your use case description of “place an order”, create a UML Sequence Diagram.

Solution

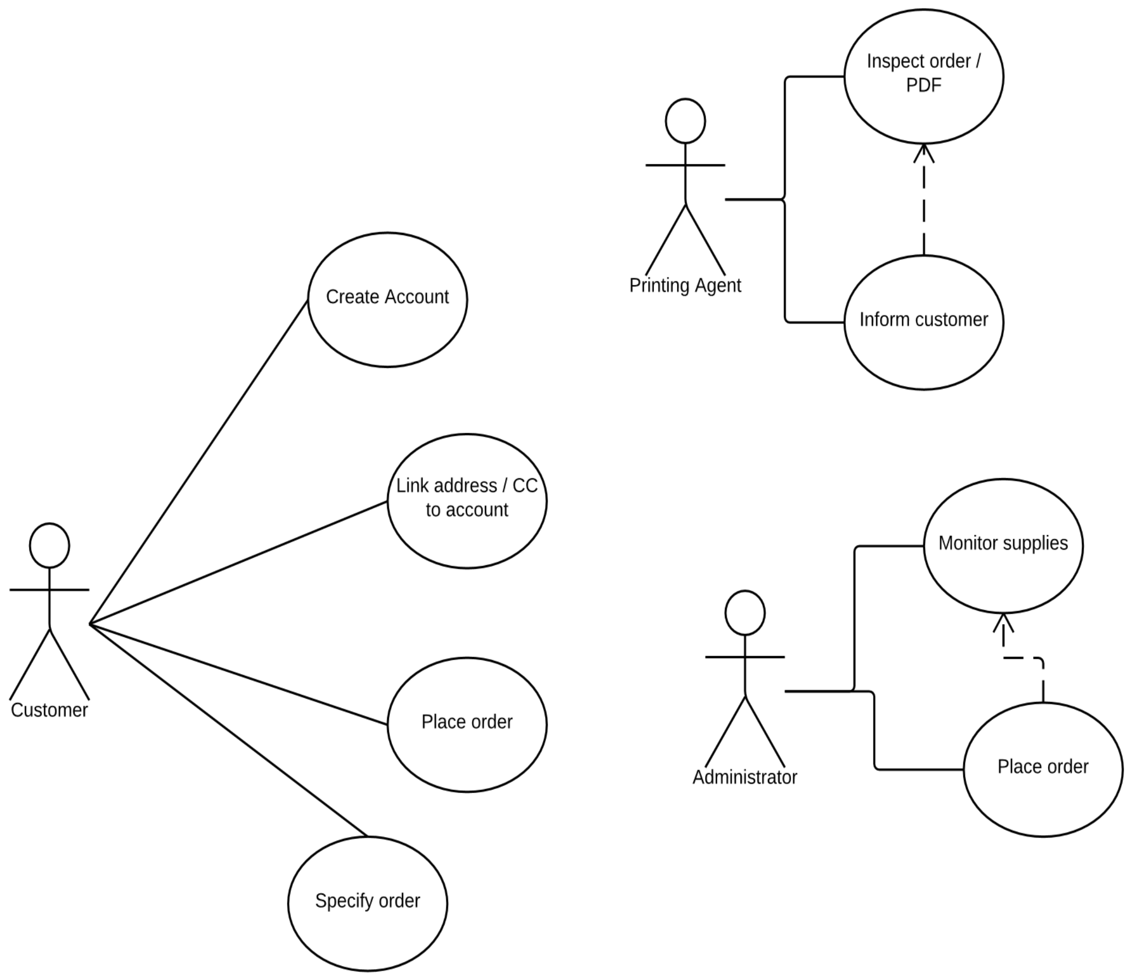
- 1 – (a)
- 2 – Figure on right – Remaining solution on next slide



Solution 2 …cont'd

- The domain categories *Student*, *Professor*, and *Seminar* are obviou candidates for classes. Professor and Student both have personal data such as name and address. This data could be factored out and placed in a superclass with a name like *Person* or *Academic*, but is not really necessary. Observing this will earn you a bonus point, whether the superclass is added or not.
- The text mentions how many professors there can be teaching a seminar, so multiplicities are apparently important. This multiplicity is required and other multiplicities should be given in the diagram as well. Although the number of seminars a professor can teach is humanly limited, a clear upperbound can not be given, so * seems appropriate here. Similarly, there may be practical or even official upperbounds to the number of seminars a student can take and the number of students that can be enrolled in a seminar, but since none was given in the text, * is the right choice here.
- The *Enrollment* class can also be expresses as an association class for the association between Student and Seminar. It is not required, but it is a straightforward way to express that each student of for each seminar has a list of grades. Alternative approaches that I have seen often have only one grade per student per seminar.
- The text is not very explicit about most of the services (methods) that should be provided, so we are lenient in this respect.

Solution 3



Place an Order

Pre: User has account
Trigger: User wants to place an order

Guarantee: Order is placed

Main:

a. Send order
b. Information is checked
c. Order is placed
d. CC is sent to bank
e. Order is finalised

Alternatives:

c-1. If information is missing, request it from the user. Retry step b.
e-1. If CC is not approved, request new CC from the user. Retry step d

Solution 3 …cont'd